BIJAN MAZAHERI

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I am interested in the information theory of causality, experimental design, and artificial intelligence. I am motivated by applications in defense and biology.

EDUCATION

California Institute of Technology - Pasadena, CA *Oct 2017 – Aug 2023*
Doctor of Philosophy
Computing and Mathematical Sciences, GPA: 3.9/4.0
Thesis Title: Combining Sources and Leveraging Contexts
Advised by Prof. Leonard Schulman and Prof. Jehoshua (Shuki) Bruck

University of Cambridge - Cambridge, UK *Oct 2016 – Jun 2017*
Classes in the Mathematical Tripos Parts IA, IB, and II.
Supported by a Herchel Smith Fellowship.

Williams College - Williamstown, MA *Sep 2012 – Jun 2016*
Bachelor of Arts
Physics and Computer Science, GPA: 3.92/4.00
Highest Honors (Physics), Phi Beta Kappa, Sigma Xi, Magna Cum Laude
Thesis Title: RNA Macrostates and Macrokinetics

ACADEMIC APPOINTMENTS

Thayer School of Engineering, Dartmouth College - Hanover, NH *Jan 2025 – Present*
Assistant Professor of Engineering
Affiliate - Quantitative and Biological Sciences
Affiliate - Computational Science and Modeling Program

Broad Institute of MIT and Harvard - Cambridge, MA
Eric and Wendy Schmidt Postdoctoral Associate *Oct 2023 – Dec 2024*
Visiting Research Scientist *Jan 2025 – Present*
Primarily advised by Prof. Caroline Uhler.

INDUSTRY EXPERIENCE

Operand – San Francisco, CA *Apr 2026 – Present*
Founding Research Advisor

Amazon Research Causality Lab - Tübingen, Germany *Oct 2022 – Feb 2023*
Intern
Worked with Dr. Michaela Hardt, Dr. Atalanti Mastakouri, and Dr. Dominik Janzing

BioDiscovery - El Segundo, CA *Jun 2017 – Sep 2017*
Intern
Worked with Dr. Soheil Shams.

IBM T.J. Watson Research Center - Yorktown Heights, NY *Jun 2016 – Sep 2016*
Intern
Worked with Dr. Victor Kravets (mentor) and Dr. Andrew Sullivan (manager).

AWARDS AND GRANTS

Dartmouth Initiative for Middle East Exchange (DIMEX) *Awarded Fall 2025*
"AISTATS 2026 and Resulting Collaborations." $22,000.

DARPA Advanced Research Concepts - COMPASS (Lead PI) *Awarded Summer 2025*
"Detecting Data Manipulation via Causal Information Geometry." $248,000.

Eric and Wendy Schmidt Postdoctoral Fellowship *Awarded Summer 2023*
1–3 years of research at the Broad Institute of MIT and Harvard. $225,000 approx.

Amazon AI4Science Research Fellowship *Awarded Spring 2022*
"Expert Graphs and Source Identification." $25,000.

National Science Foundation Graduate Research Fellowship *Awarded Spring 2019*
3-year Ph.D. fellowship. $138,000 approx.

Herchel Smith Fellowship *Awarded Spring 2016*
Funding for 1–3 years of study at the University of Cambridge. $100,000 approx.

TEACHING

Dartmouth Engineering
Discrete and Probabilistic Systems *Summer 2025*
Principles of Causality *Spring 2025, 2026*

Teaching Assistant
Markov Chain Monte Carlo (Caltech) *Spring 2022*
Discrete Mathematics (Williams) *Fall 2015*
Electricity and Magnetism (Williams) *Spring 2015*
Introduction to Mechanics (Williams) *Fall 2013*

PUBLICATIONS

Bijan Mazaheri, Jiaqi Zhang, and Caroline Uhler. Relaxing Faithfulness for Intervention-Only Causal Discovery. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.

Bijan Mazaheri, Jiaqi Zhang, and Caroline Uhler. Meta-Dependence in Conditional Independence Testing. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.

Bijan Mazaheri, Spencer Gordon, Yuval Rabani, and Leonard J. Schulman. Causal Discovery under Latent Class Confounding. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2026.

Bijan Mazaheri, Chandler Squires, and Caroline Uhler. Synthetic Potential Outcomes and Causal Mixture Identifiability. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.

Bijan Mazaheri, Siddharth Jain, Matthew Cook, and Jehoshua Bruck. Omitted Labels Induce Nontransitive Paradoxes in Causality. In *Conference on Causal Learning and Reasoning (CLeaR)*, 2025.

Spencer Gordon, Erik Jahn, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. Identification of Mixtures of Discrete Product Distributions in Near-Optimal Sample and Time Complexity. In *Conference on Learning Theory (COLT)*, 2024.

Bijan Mazaheri, Atalanti Mastakouri, Dominik Janzing, and Michaela Hardt. Causal Information Splitting: Engineering Proxy Features for Robustness to Distribution Shifts. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.

Spencer Gordon, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. Causal Inference Despite Limited Global Confounding via Mixture Models. In *Conference on Causal Learning and Reasoning (CLeaR)*, 2023.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Glioblastoma Signature in the DNA of Blood-Derived Cells. *PLOS ONE*, 16(9): e0256831, 2021.

Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. Expert Graphs: Synthesizing New Expertise via Collaboration. In *IEEE International Symposium on Information Theory (ISIT)*, 2021.

Spencer Gordon, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. Source Identification for Mixtures of Product Distributions. In *Conference on Learning Theory (COLT)*, 2021.

Bijan Mazaheri, Siddharth Jain, and Jehoshua Bruck. Robust Correction of Sampling Bias Using Cumulative Distribution Functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

* Authorship order is alphabetical.

PREPRINTS

Zou Yang, Sophia Xiao, and **Bijan Mazaheri**. Masking Causality and Conditional Dependence. Under review, 2026.

Liyuan Xu and **Bijan Mazaheri**. Estimating Aleatoric Uncertainty in the Causal Treatment Effect. Under review, 2026.

Mateusz Gajewski, Sophia Xiao, and **Bijan Mazaheri**. Data Augmentation via Causal Residual Bootstrapping. Under review, 2026.

Spencer Gordon, **Bijan Mazaheri***, Yuval Rabani, and Leonard J. Schulman. The Sparse Hausdorff Moment Problem, with Application to Topic Models. *arXiv:2007.08101*, 2020.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Cancer Classification from Healthy DNA using Machine Learning. *bioRxiv:517839*, 2019.

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Short Tandem Repeats Information in TCGA is Statistically Biased by Amplification. *bioRxiv:518878*, 2019.

* Authorship order is alphabetical.

PATENTS

Siddharth Jain, **Bijan Mazaheri**, Netanel Raviv, and Jehoshua Bruck. Mutation Profile and Related Labeled Genomic Components, Methods and Systems. US Patent App. 20190392951A1, 2019.

INVITED TALKS

CauScale Spotlight @ AISTATS *May 2026*
"Masked Unfairness: Hiding Causality in Zero ATE"

Tsinghua Sanya International Mathematics Forum *Jan 2026*
"Distribution-Level Techniques for Heterogeneous Causal Relationships"

Biomedical Data Science Seminar Series, Dartmouth Medical School *Oct 2025*
"Causal Underpinnings of Information Synthesis"

Williams College Statistics Colloquium *Oct 2025*
"Causal Underpinnings of Information Synthesis"

Ops/MS Brown Bag Seminar, Tuck School of Business *Feb 2025*
"Synthetic Potential Outcomes and Causal Mixture Identifiability"

Boston University Machine Learning Symposium *Nov 2024*
"Synthetic Potential Outcomes and Causal Mixture Identifiability"

Stanford Online Causal Inference Seminar *Oct 2024*
"Synthetic Potential Outcomes and the Hierarchy of Causal Identifiability"

Jones Seminar, Thayer School of Engineering at Dartmouth *May 2024*
"Latency and Heterogeneity in Data and What to Do About It"

Simons Institute for the Theory of Computing *May 2023*
"Causal Discovery under Limited Global Confounding"

RESEARCH MENTORSHIP

PhD Advisees
Hamza Virk (Thayer School of Engineering, Dartmouth)
Zou Yang (Thayer School of Engineering, Dartmouth)
Sophia Xiao (Thayer School of Engineering, Dartmouth)

Graduate Students
Mateusz Gajewski (PhD Student, Poznań University of Technology)
Ryan Montgomery (MBA Student, Tuck School of Business, Dartmouth)

Undergraduate Students
Haley Lin (Williams College)
Rafael Castro (California Institute of Technology SURF)
Youmi Ji (Dartmouth College URAD)
Keion Grieve (Dartmouth College URAD)
Caroline Krantz (Dartmouth College URAD)
Muhammad Ahmad (Dartmouth College FYREE)
Hailey King (Dartmouth College)
Ruchita Nair (Dartmouth College)
Benjamin Cavanagh (Dartmouth College)
Sreshth Tiwari (Dartmouth College URAD)
Connor Kilkenny (Dartmouth College URAD)
Titus Johnson (Dartmouth College URAD)
Tamier Baoyin (B.A. Mt. Holyoke College, B.E. Dartmouth College)
Leila Salken (Williams College)

Anika Roy (B.S. IIIT Hyderabad, Khorana Scholar, Heidelberg Future Leaders Award)
Beshr Bouli (MIT UROP)

Data Science Projects
I have supported over 30 projects with undergraduate students using data on my website, and have advised some of these students in applying to graduate school.

WORKSHOPS

CauScale @ AISTATS *May 2026*
"Masked Unfairness: Hiding Causality in Zero ATE"

Tsinghua Sanya International Mathematics Forum *Jan 2026*
"Causality and Machine Learning"

CauScien Workshop at NeurIPS *Dec 2025*
Attendee.

Scaling Up Intervention Models at ICML *Jul 2025*
"Faithfulness and Intervention-Only Causal Discovery"

Simons Institute for the Theory of Computing *Spring 2022*
"Causality"

SERVICE

Conferences
CLeaR: Reviewer (2024–2026), Program Chair (2026).
NeurIPS: Reviewer (2020–2025), Top Reviewer Award (2025).
UAI: Reviewer (2026).
AISTATS: Reviewer (2023–2026).
ICML: Reviewer (2025).

Journals
Nature Machine Intelligence: Reviewer (2024)
The American Statistician: Reviewer (2024)
Electronic Journal of Statistics: Reviewer (2025)

Committees
Dartmouth Engineering 3-2 and 2-1-1-1 Programs
Curriculum, Dartmouth Computational Science and Modeling Program

Other Service
Faculty Advisor, Dartmouth Cross Country and Track and Field Teams
Volunteer Assistant Coach, Caltech Cross Country and Track and Field Teams
Sports statistics outreach for Cross Country and a correspondent at D3 Glory Days

PROJECTS

LACCTiC *Sep 2021 – Present*
I maintain a website for collegiate cross country with 10,000 regular users that applies concepts from batch-effect correction to ranking performances on differing terrain. The backend runs on Python and Django and the frontend uses React, and the database is hosted on AWS. I have helped advise over 30 student projects using this data.